FREIGHT RATE PREDICTION

Machine Learning Assessment Report

Project: Freight Rate Prediction Challenge
Prediction target: Posted freight rate
Training period: January–August 2025
Validation period: September–October 2025

Executive Summary

This project develops a supervised machine learning pipeline to estimate freight load rates. The training dataset contains 48,000 records covering January through October 2025. Two tree-based regression models, Random Forest and Extra Trees, were evaluated using a chronological validation strategy. Extra Trees achieved the best validation performance, with a mean absolute error (MAE) of $150.82 and an R-squared value of 0.8175. The selected model was subsequently used to generate predictions for the 12,000 validation loads and 31 December scenario records.

1. Problem Definition and Dataset

The objective is to predict the posted freight rate using available shipment characteristics. The training dataset contains 48,000 observations and 14 columns, including origin and destination, shipment distance, equipment type, weight, date, market indicators, and the target variable *posted_rate*.

The target variable has a mean of $2,373.98, a median of $2,030.76, and a standard deviation of $1,486.49. Its distribution is right-skewed, with a small number of high-value observations.

2. Exploratory Data Analysis

Exploratory analysis identified several relevant patterns:

Distance: Distance has a strong positive correlation with posted rate (r = 0.9085), making it an important predictive feature.

Equipment: Reefer loads have the highest average posted rate ($2,553.64), followed by Flatbed ($2,445.09) and Dry Van ($2,271.55).

Weight: Weight has a weak linear correlation with posted rate (r = 0.0348). It was retained as a candidate feature because tree-based models can capture nonlinear relationships.

Data quality: Weight contains 300 missing values and 292 negative values. Market index contains 374 missing values. The dataset contains no duplicate rows.

The interquartile-range analysis identified 260 potential target outliers (0.54%). These observations were retained because high rates may represent legitimate long-distance shipments.

3. Preprocessing and Feature Engineering

Pickup, delivery, and equipment were treated as categorical features. Distance, weight, and date-derived variables were treated as numerical features. The date column was transformed into year, month, day, day of week, and day of year.

Negative shipment weights were treated as missing values. Median imputation was applied to numerical features, while categorical missing values were imputed using the most frequent category. Standard scaling was applied to numerical features, and one-hot encoding was used for categorical variables. Unknown categories were handled using *handle_unknown='ignore'*.

The final model uses features available in both the training and December input datasets. Market index, quote signal, and geographical coordinates were excluded because the December inputs do not provide them.

4. Training and Validation Strategy

A chronological split was used to approximate deployment on future freight loads. Records dated before 1 September 2025 were assigned to training, while September and October records were reserved for validation.

Dataset	Date range	Records
Training	Jan–Aug 2025	38,477
Validation	Sep–Oct 2025	9,523

Both models used 80 estimators, a maximum depth of 20, a minimum of five samples per leaf, and random_state=42. The models were trained through preprocessing and regression pipelines to keep preprocessing fitted on training data only.

5. Model Evaluation and Selection

The models were compared using MAE, root mean squared error (RMSE), and the coefficient of determination (R-squared). MAE was selected as the primary model-selection criterion.

Model	MAE ($)	RMSE ($)	R^2
Random Forest	172.33	658.04	0.8141
Extra Trees	150.82	651.87	0.8175

Extra Trees achieved the lowest MAE and RMSE and the highest R-squared value. It was therefore selected as the final model. Its validation MAE of $150.82 represents the average absolute difference between predicted and actual posted rates on the chronological validation set.

6. Prediction Outputs

The selected Extra Trees model generated predictions for all 12,000 validation records. The predictions were written to *validation_predictions.csv* using the supplied template structure and load identifiers.

The model also generated predictions for 31 December 2025 scenario records. These records represent the same route (Lexington to Fort Wayne), distance (360 miles), equipment (Dry Van), and weight (32,000 lb), with the date varying across the month.

Statistic	Predicted rate ($)
Mean	859.66
Standard deviation	3.39
Minimum	854.57
Median	859.86
Maximum	866.99

The predictions range from $854.57 to $866.99. Their relatively small variation reflects the fact that shipment characteristics remain constant while the date changes.

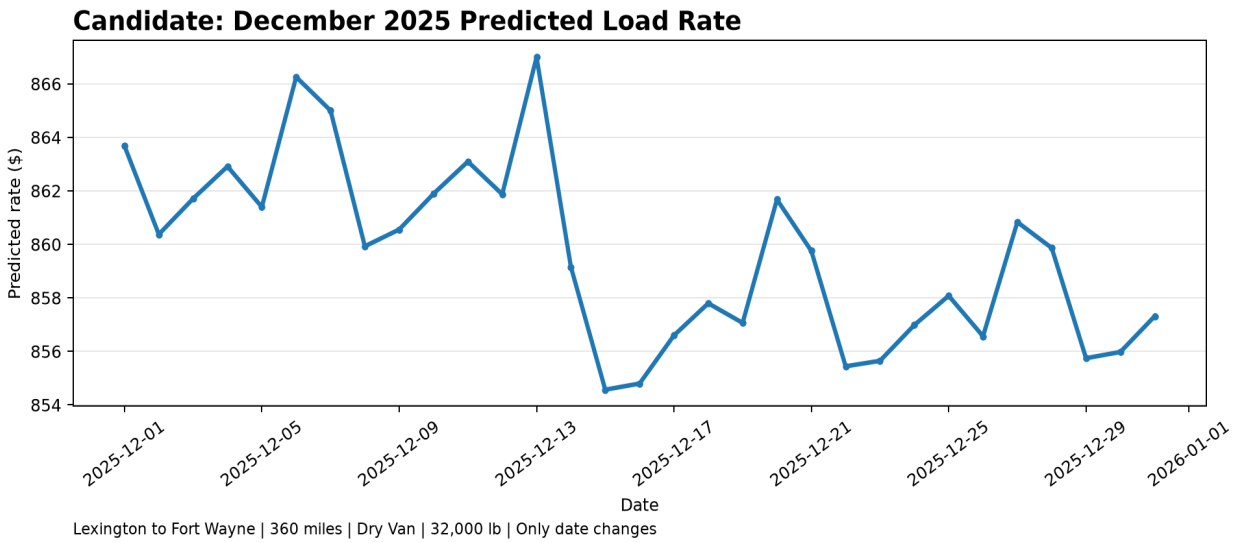


Figure 1. December 2025 predicted rates for the Lexington–Fort Wayne shipment scenario.

7. Limitations

The validation results measure performance on September–October 2025 records and may not fully represent performance under future market conditions. Market index, quote signal, and coordinates were excluded to maintain feature consistency with the December inputs. The December scenario contains only one route and should not be interpreted as a general forecast of market-wide freight rates.

The local output-check script verifies prediction-file structure and validity and generates the December chart. It is a reconstructed utility, not a verified official Spotter scoring implementation.

Conclusion

The project produced a reproducible freight-rate prediction pipeline using chronological validation and tree-based regression. Extra Trees was the strongest of the two evaluated models, achieving a validation MAE of $150.82. The final pipeline generated complete validation and December prediction files. Further improvement could involve route-level features, additional temporal data, and systematic hyperparameter tuning.